AUTOSAR Classic Platform — Door Control System

Synthetic HLD Test Document for RAG / Entity Extraction Validation

Document ID: AUTO-HLD-DC-001 **Revision:** 1.0 **Status:** Test Baseline

Purpose: This synthetic document is designed specifically to test an engineering RAG system. It contains explicit software components, ports, interfaces, requirements, architecture relationships, timing information, and deliberately absent facts for negative-grounding tests.

1. System Overview

The Door Control System (DCS) is an automotive body-control application responsible for processing lock and unlock requests and publishing door status information. The example uses an AUTOSAR Classic Platform-style application architecture. The application software is separated from Basic Software (BSW) through the Runtime Environment (RTE). Software components communicate through ports and port interfaces.

The system contains three application software components: DoorControlSWC, CentralLockSWC, and DiagnosticSWC. CentralLockSWC provides lock and unlock requests to DoorControlSWC. DoorControlSWC processes these requests and publishes the resulting door status. DiagnosticSWC receives diagnostic requests and publishes diagnostic responses.

2. AUTOSAR Layered Architecture Context

Layer	Responsibility in this test system
Application Software	Contains DoorControlSWC, CentralLockSWC, and DiagnosticSWC.
RTE	Provides the communication boundary between application software components and the underlying BSW.
Basic Software (BSW)	Provides platform services and hardware-related infrastructure below the RTE.

3. Software Components

Software Component	Type	Responsibility
DoorControlSWC	Application SWC	Processes door lock/unlock commands and publishes door status.
CentralLockSWC	Application SWC	Generates lock and unlock requests for the door control function.
DiagnosticSWC	Application SWC	Processes diagnostic requests and provides diagnostic responses.

4. Ports

SWC	Port	Direction	Interface
CentralLockSWC	P_LockRequest	Provided	LockRequestInterface
DoorControlSWC	R_LockRequest	Required	LockRequestInterface
DoorControlSWC	P_DoorStatus	Provided	DoorStatusInterface
DiagnosticSWC	R_DiagnosticRequest	Required	DiagnosticRequestInterface
DiagnosticSWC	P_DiagnosticResponse	Provided	DiagnosticResponseInterface

5. Port Interfaces and Data Elements

Port Interface	Data Elements	Description
LockRequestInterface	LockCommand	Boolean command indicating a lock request.
LockRequestInterface	UnlockCommand	Boolean command indicating an unlock request.
DoorStatusInterface	DoorOpen	Indicates that the door is open.
DoorStatusInterface	DoorClosed	Indicates that the door is closed.

Port Interface	Data Elements	Description
DiagnosticRequestInterface	DiagnosticCommand	Diagnostic command received by DiagnosticSWC.
DiagnosticResponseInterface	DiagnosticStatus	Result status returned by DiagnosticSWC.

6. Architecture and Communication Flow

Flow 1 — Lock/Unlock: CentralLockSWC provides P_LockRequest. The request uses LockRequestInterface and is consumed by DoorControlSWC through R_LockRequest. DoorControlSWC evaluates LockCommand or UnlockCommand and then publishes the resulting state through P_DoorStatus using DoorStatusInterface.

Flow 2 — Diagnostics: DiagnosticSWC receives diagnostic commands through R_DiagnosticRequest using DiagnosticRequestInterface. After processing the command, DiagnosticSWC publishes the result through P_DiagnosticResponse using DiagnosticResponseInterface.

System relationship: CentralLockSWC → LockRequestInterface → DoorControlSWC → DoorStatusInterface. DiagnosticSWC is logically independent of the lock request path and provides a separate diagnostic path.

7. Requirements

ID	Requirement
DCS-REQ-001	DoorControlSWC shall process valid lock requests received through R_LockRequest.
DCS-REQ-002	DoorControlSWC shall process valid unlock requests received through R_LockRequest.
DCS-REQ-003	DoorControlSWC shall publish the resulting door state through P_DoorStatus.
DCS-REQ-004	CentralLockSWC shall provide lock and unlock commands through P_LockRequest.
DCS-REQ-005	DiagnosticSWC shall accept diagnostic commands through R_DiagnosticRequest.
DCS-REQ-006	DiagnosticSWC shall publish a diagnostic result through P_DiagnosticResponse.

8. Timing and Execution

For this test baseline, DoorControlSWC contains a DoorCommandRunnable configured with a nominal 10 ms periodic execution cycle. The runnable evaluates the latest lock/unlock command and updates the door state output. DiagnosticSWC contains DiagnosticRunnable with a nominal 50 ms periodic execution cycle.

Important: The timing values above are project-specific assumptions for this synthetic HLD and are not presented as AUTOSAR standard requirements.

9. ECU Mapping

ECU	Mapped Software Components	Network
BodyControlECU	DoorControlSWC, CentralLockSWC, DiagnosticSWC	CAN-FD

10. Safety and Diagnostics Notes

The synthetic design assumes that invalid or unavailable lock commands are rejected by DoorControlSWC and reported through the diagnostic path. A production implementation would require project-specific safety analysis, diagnostic specifications, timing analysis, and verification evidence. This document does not assign an ASIL and does not define a production safety case.

11. Explicitly Unspecified Information

The following information is intentionally NOT defined in this document: microcontroller vendor and model, CPU frequency, exact RTOS implementation, AUTOSAR vendor/toolchain, compiler version, memory addresses, CAN IDs, cryptographic algorithms, ASIL classification, and production calibration values. A grounded RAG system should not invent these values when asked about them.

12. Glossary

Term	Meaning in this document
SWC	Software Component.
RTE	Runtime Environment connecting application software to the AUTOSAR infrastructure.
BSW	Basic Software.

Term	Meaning in this document
Port	Communication endpoint of a software component.
Port Interface	Defines the communication/data contract associated with ports.
VFB	Virtual Functional Bus concept used to decouple application software from infrastructure.

This is a synthetic engineering test artifact created for validation of the AUTOSAR RAG prototype. It is not an official AUTOSAR specification or AUTOSAR release document.